1 How the SMOTE Literature Uses Data

A reader who wants to compare two SMOTE variants, or to choose a benchmark for a new one, needs answers to questions that the papers rarely answer together. Can the numbers that two papers report on a dataset of the same name be compared? How broad is the evidence behind the benchmarks the field relies on? How difficult, beyond the imbalance ratio, are the datasets that can be checked? When a paper reports that its method beats SMOTE, on what kind of data does that happen, and can a reader check it? This section answers these questions from the 1,654 dataset entries and the comparisons with SMOTE that we extracted from the 76 papers of our corpus: 835 under one fixed metric per paper (834 per dataset and one average over datasets) and 2,497 over every metric the papers report. Three of them (ENN, AdaBoost and one review) contain no evaluation on data, so counts below use the 73 papers with an evaluation on data; Tomek links is kept because it illustrates its method on one synthetic two-class set.

The central observation is that the evidence for a new oversampler is only as interpretable as the description of its data, and in this literature that description is often incomplete: the same name can denote different tasks, many modifications are not described well enough to be repeated, and 14 of the 57 papers that use SMOTE as a baseline give no per-dataset pair of scores against it. Where per-dataset results exist, the proposed method beats plain SMOTE in about four of five decided comparisons. These wins are not more frequent on data the authors modified, but they are more frequent on data whose minority examples are borderline or isolated, and they are much less consistent on precision and specificity than on the other metrics.

1.1 Extraction Procedure

For every paper we read the full text and any supplementary file in our collection, and recorded each dataset as printed, including spelling, with its group in the paper, source repository, size, number of features and classes, minority and majority counts, and the imbalance ratio where the paper prints them, together with the table and page they come from. For each dataset we also recorded how it reached the experiment: *original* (used as distributed, with its native classes), *repository-defined* (a two-class split published by a repository such as KEEL or imbalanced-learn, used as is), *modified* (the authors chose the minority classes, merged or dropped classes, set a threshold, undersampled or injected noise), or *synthetic* (generated by the authors). Where a paper did not state its source, two fixed rules were applied and flagged: names that follow the KEEL file convention (for example `ecoli-0-1_vs_5`, `yeast4`) were labelled repository-defined, and common names in multiclass papers with no stated change were labelled original. Two further rules were fixed before counting: a set that is two-class in its source and is used without comment was labelled original, and a set from which the authors removed rows (for example, records with missing values) or attributes was labelled modified.

For the results we copied, per dataset, the printed score of the proposed method and of SMOTE from one results table selected by a fixed rule: G-mean, else AUC, else F-measure, else accuracy; the classifier highlighted in the abstract, else the first reported; the unperturbed condition. We recorded values rather than wins so that any definition of a win can be applied later. From the same table setting we then copied every other metric that the paper reports for both methods (2,497 dataset-by-metric values in 44 papers); the fixed-metric values were copied a second time in this pass, independently, and agree on all 839 rows that both copies contain. Plain SMOTE was distinguished from SMOTE embedded in a multiclass or ensemble wrapper (Static-SMOTE, OVO-SMOTE [FERNANDEZ2013ANALYSING], SMOTE-all, SMOTE+AdaBoost). Values printed in a paper are never overwritten; corrections against the source repository are stored beside them

with their justification. Every extracted value was then checked against the PDF a second time: a value was accepted when the dataset name and its numbers appear on the same line of the cited page, and every table that is printed as an image or whose text layer is garbled was read from the rendered page instead. All 1,654 dataset entries and 835 fixed-metric comparisons passed this check; the values of the other metrics were read once, from the text layer or the rendered page.

Evidence: ai-draft/full/SCHEMA.md; build/full_extraction.py; KEEL and multiclass rules approved by Toon 18 Oct, rules A and B 18 Oct; 454 extraction notes in out/full_uncertain.csv, each resolved in out/uncertain_resolution.csv; second check: build/verify_pdf.py, out/visual_checks.json, out/verified_*.csv; values also checked by Toon against the PDFs (28 Oct 2026).

1.2 The Benchmark Landscape

The 73 papers with an evaluation on data list 1,654 dataset entries. One of them, the single toy example in Tomek links, is a description rather than a name and is set aside here, leaving 1,653 named entries. These reduce to 771 distinct names once names are lower-cased, text in parentheses is dropped, and underscores and spaces are treated alike. Treating spellings that differ only in punctuation as one dataset (`page-blocks`, `pageblocks` and the singular `page-block`) leaves 680 datasets, and grouping each numbered or split version with its main set (for example `yeast4` or `glass-0-1-6_vs_2` with yeast and glass), while keeping non-standard names such as `glass identification` apart, leaves 459 source families. Most datasets appear once: 463 of the 680 occur in a single paper. Counted as datasets, ecoli appears in 29 papers, yeast in 22 and glass in 19; counted with their splits, the ecoli family appears in 42 papers, the yeast family in 39 and the glass family in 35, about half of the 73. Half of all uses of a family fall on 37 of the 459 families. Names whose printed number of features contradicts their main set are kept in its family when they are the same source with a different composition (for example `abalone8`, `oil1`) and kept apart when they are a different data set that shares the prefix (for example `car1`, a microarray set with 9,182 features).

Splits cut from one source are not copies of each other. Among the verifiable two-class KEEL files (Section 1.5), the 14 yeast splits range in imbalance ratio from 2.5 to 41.4 and in noise index from 0.29 to 0.86, and the 19 ecoli and 13 glass splits from 0.12 to 0.69 and from 0.24 to 0.77. They do share the source's attributes (7 or 8 for yeast, 6 or 7 for ecoli, 9 for glass), its domain and often its examples: `yeast4` and `yeast-0-5-6-7-9_vs_4` use the same 51 minority examples and differ only in which classes form the majority.

Outcomes on splits of one source are also related. In the eleven papers that compare with plain SMOTE on several splits of the same source (282 per-dataset comparisons, 262 same-family pairs; families by the rule in the evidence note), whether the proposed method wins does not cluster by family (pairs agree on win, tie or loss in 0.60 of cases, as under random relabelling), but the size of its margin does: the within-paper ranks of the margin differ by 0.29 on average for two splits of one source against 0.34 when family labels are permuted within each paper ($p = 0.005$, 10,000 permutations; leaving out one paper at a time gives p between 0.002 and 0.048).

Splits cut from one source vary the task (its imbalance ratio and where the minority examples lie) but not the source (its domain, attributes and dimensionality), and within the papers that evaluate several splits of one source, the margin over SMOTE tends to be slightly more alike within a family than between families. Papers should therefore report the number of source families alongside the number of datasets, and should not treat each split as independent evidence.

What the reader should do. Report the source family of each dataset alongside the count, so that readers can see how many independent sources stand behind a claim, and keep splits of one

source together when testing for significance.

Evidence: out/verified_datasets.csv; counts reproduced by build/reconcile_counts.py (out/reconcile_counts.json): rule R1 sets aside the Tomek links placeholder (row 1558, kept in the file), R2 normalises with build/analyse_dataset_usage.py norm(), R3 gives exact-name, dataset and family counts under rule T1 (build/family_rule.py, out/dataset_families.csv; the earlier prefix count gave 39/39/36 and is superseded); datasets and families per name: out/landscape_families.csv; concentration by dataset and family: out/benchmark_concentration.json. With the placeholder the counts are 772 and 547; removing all punctuation would leave 678 names. The 605 names of the 6 Oct list are superseded by the 771 of the full re-read. Spread within the yeast, ecoli and glass families: difficulty/family_spread.py → family_spread.json. Datasets and families everywhere in this subsection: rule T1 (Toon, 28 Oct 2026; names differing only by punctuation are one dataset, numbered or split versions belong to their main set, non-standard names are not merged), build/family_rule.py → out/dataset_families.csv; outcomes linked to records through out/outcome_dataset_links.csv; build/family_dependence.py → out/family_dependence.json (key t1).

Candidate benchmarks for a cross-family comparison. Table 1 lists 18 datasets considered as benchmarks for comparing method families: the 11 datasets we set out to examine and the datasets that meet one of two written rules. The first rule takes every dataset used by at least ten papers (33 datasets). The second takes datasets used by three to nine papers whose every printed size is at least 10,000 instances, a size that no entry of the first group reaches (7 datasets). The groups follow from these two rules on use and printed size only and say nothing about overlap, noise, sparsity, feature type or any other property. IR is not used, because its definition differs between papers. No dataset used by three or more papers is printed with 100 or more features every time it is reported; this is a property of the rule, not evidence that the corpus contains no high-dimensional data sets.

Family counts follow the same rule for every dataset. The dataset `shuttle` is printed with four different sizes (2,175, 14,500, 43,500 and 57,999 instances) and once without a size. Two pairs of papers print matching sizes and class counts; the other versions are single reports, and OVO-SMOTE prints five classes in one table and seven class counts in another. The versions are therefore kept apart and not pooled.

The frequently used benchmarks are small: among the entries that print a size, half have 800 or fewer instances and half have nine or fewer features. Names printed with 10,000 or more instances in every paper are used by three to nine papers each, so a comparison across method families that wants to cover data size has to look beyond the frequent set.

Limits of the evidence. Size, dimensionality and the number of classes are printed for 80.2%, 75.9% and 77.4% of the 1,653 named entries, and class proportions in some form for 56.8%. A measured class overlap exists for 36 entries in two papers, on two incompatible scales, and the curated evidence contains no statement about sparsity. What the papers print therefore cannot show which ranges of overlap, noise or sparsity the benchmarks cover; the candidates in Table 1 are chosen on use and size only. Section 1.5 measures these properties directly for the entries whose files can be identified.

Evidence: out/benchmark_candidates.csv and .json (build/landscape_candidates.py; rules and approval status in `criteria`); table rows written by build/landscape_manuscript.py; out/landscape_families.csv (446 groups, 82 flagged and left unmerged); out/shuttle_versions.csv (build/shuttle_versions.py; OVO-SMOTE Table 2 and Table 3, p.6, checked 19 Oct); out/characteristics_coverage.json; out/benchmark_concentration.json. All records are browsable in the Atlas (manuscript tab and Landscape tab).

Table 1: Data Sets Considered as Benchmarks for Cross-Family Comparison. F: used by ten or more papers; D: used by three to nine papers, every printed size at least 10,000 instances; V: needs verification. Papers count the dataset (spellings that differ only in punctuation are one dataset); family papers include the numbered and split versions of the dataset. Values are as printed and differ between papers. [a]Four printed versions, not pooled.

Dataset	Group	Papers	Family papers	Instances	Size printed	Classes	Modified
ecoli	F	29	42	301–336	25/31	2, 4, 5, 8	8/31
page-blocks	F	22	34	545–5,476	17/22	2, 4, 5	6/22
yeast	F	22	39	483–1,485	16/22	2, 6, 8, 9, 10	4/22
pima	F	21	21	590–769	19/23	2	2/23
vehicle	F	21	30	713–940	19/23	2, 3, 4	10/23
haberman	F	20	20	252–306	19/22	2	2/22
glass	F	19	35	167–214	18/21	2, 6	7/21
balance	F	16	16	625	14/16	3	3/16
wine	F	16	16	130–178	15/18	2, 3	6/18
satimage	F	15	16	6,430–6,435	11/15	2, 6, 7	8/15
letter	D	9	9	11,057–20,000	9/9	2, 26	6/9
nursery	D	7	7	12,690–12,960	7/7	2, 4, 5	3/7
mammography	D	5	5	11,183	4/5	2	0/5
adult	D	3	3	32,562–48,842	2/3	2	1/3
avila	D	3	4	10,430–20,850	3/3	2, 11	1/3
default	D	3	3	30,000	2/3	2	0/3
shoppers	D	3	3	12,330–12,331	3/3	2	0/3
shuttle	V	7	15	2,175–57,999[a]	6/7	2, 5	1/7

1.3 How Datasets Are Chosen and Modified

Table 2 shows that modification by the authors is common: 331 of 1,654 dataset entries were changed by the authors themselves, and for 199 the paper does not let a reader tell how the data were obtained. Eleven papers give no reason for their choice of datasets or only call them widely used.

Modification is not a fault in itself: binarising a multiclass set or raising the imbalance ratio are legitimate ways to probe a method. What matters for the reader is whether the modification is declared where a reader looks for it. A KEEL file name such as `yeast-0-5-6-7-9_vs_4` declares its split; a paper that prints only "Yeast" after choosing its own minority classes does not.

Modifications differ in how completely they are described. Of the 331 modified entries (39 papers), 177 in 30 papers change the class structure, most often by keeping one class against the rest (84 entries), and 147 in 9 papers raise the imbalance ratio on purpose, almost always by removing minority examples at random (131 entries). A purpose for the change is stated in 15 of the 39 papers. For 93 of the 331 entries, information needed to repeat the change is missing: the characteristics of the modified set are not printed for 55, the target imbalance ratio for 18 and the minority class for 15. At the paper level, 8 of the 37 papers that binarise data do not state which class becomes the minority. Entries modified without printed characteristics cannot be matched to any public file, and 28 of them are therefore left out of the difficulty measurement (Section 1.5).

Evidence: out/data_handling_summary.json (paper-level codes; binarised_without_stated_rule, minority_rule_among_binarised); out/full_summary.json how_obtained_all (entry level, with rules); operations, purposes and missing information: build/modified_operations.py → out/modified_operations_audit.csv and out/modified_operations_summary.json (Analysis A).

Table 2: How the Papers Choose and Obtain Their Data. Top: paper level, 73 papers with an evaluation on data. Bottom: dataset entries.

Reason given for the choice of datasets (papers)	
Own reason (diversity, difficulty, application, ...)	48
Cites prior work	14
"Widely used" only	3
Not stated	8
Benchmark list adopted from earlier work, declared (papers)	20 (wholesale 11)
Binarised, minority rule not stated (papers)	8 of 37
How each dataset entry reached the experiment	
Original	534
Repository-defined (KEEL, imbalanced-learn)	470
Modified by the authors	331
Synthetic	120
Cannot be told from the paper	199

1.4 The Same Name Is Not the Same Data

Of the 192 datasets that at least two papers print with a size (spellings that differ only in punctuation counted as one dataset), 90 appear with more than one size: thyroid with 7,200 instances in five papers and 720 in four, dermatology with 358 (the KEEL version) in eight and 366 (UCI) in two, ecoli with 336, 327, 310 and 301, and pima with 768, 769, 635 and 590. Some of these differences are declared subsamples; the name alone does not tell which. MWMOTE [BARUA2014MWMOTE] illustrates the problem inside a single paper: its Satimage (6,435 instances) and Statlandsat (4,435) are the full Statlog Landsat set and its training file, counted as two of its twenty datasets, and its common names are ambiguous in both directions (its Ecoli matches the KEEL file `ecoli1` exactly, while its Yeast matches no published split).

Imbalance ratios add a second layer. Table 3 lists the definitions found in the corpus; some are not convertible into one another, and some papers print IR without a formula.

Errors also travel with shared benchmark tables: MC-CCR and MC-RBO both print an IR of 1.17 for vehicle, a value no definition produces from its class counts.

What the reader should do. Compare results across papers only after mapping each name to its exact file and class split, and report the IR formula together with its values.

Evidence: out/verified_datasets.csv: names normalised, printed n_instances compared across papers (192 datasets, rule T1, with a printed size in two or more papers; 198 if spellings are kept apart; out/reconcile_counts.json size_variation); the earlier 20 of 75 from ai-draft/dataset_variants.json covered 8 papers only and is superseded; mwmote-table5-audit.md; extraction-notes.md; full re-read ir_definition per paper in out/full_summary.json.

1.5 How Difficult Are the Datasets That Can Be Checked?

The papers rarely describe how hard their data are beyond the imbalance ratio. A numerical overlap measure is printed for 36 of the 1,653 named entries, all in two papers, and noise or borderline examples are described for 5; three of 72 papers report such a property of their original data (Section 1.2). We therefore measured difficulty ourselves on the entries whose data can be identified. An entry counts as verifiable when its name matches a public file and the printed number of instances (and of minority examples, when printed) equals that file's, or when the paper prints no size but the name matches a KEEL file and the authors did not modify the data. This holds for

Table 3: Imbalance-Ratio Definitions Printed in the Papers

Definition	Formula	Used by (examples)
Largest over smallest class	$N_{\max}/N_{\min}$	MDO [ABDI2016MDO], CBH-SID [PALLI2022CBHSID], MKC-SMOTE [WANG2024MKC] (its IR)
Majority over minority (binary wording)	$N_{\mathrm{maj}}/N_{\min}$	OVO-SMOTE (defined for the binary case, applied to multiclass sets)
Percentage minority	$100\,N_{\min}/N$	RUSBoost [SEIFFERT2010RUSBOOST]
Per-class imbalance degree	$\sum_{q\neq i} N_q/((C-1)N_i)$	OREM-M [ZHU2023OREM] (overall IR only cited, not printed)
Summed pairwise multiclass IR	$\sum_{i<j}(N_i/N_j - 1)$	SMOM [ZHU2017SMOM], GDHS [YAN2025GDHS], MKC-SMOTE (IR_{overall})
Imbalance degree Id	distance from uniform [ORTIGOSAHERNANDEZ2017IMBALANCE]	SHSampler [ZHENG2022SHSAMPLER] (cited, no formula printed)
Minimum class prior	$N_{\min}/N$	DSRBF [FERNANDEZNAVARRO2011DSRBF]
Coefficient of variation	$\mathrm{sd}(N_i)/\mathrm{mean}(N_i)$	COG-OS [WU2007COGOS]
Proportion pair	e.g. 0.04:0.96	MWMOTE (wording: minority to majority)
Not stated (values equal $N_{\max}/N_{\min}$)	n/a	MC-CCR [KOZIARSKI2020MCCCR], MC-RBO [KRAWCZYK2020MCRBO]

643 entries from 61 papers, which use 131 files. Entries modified by their authors without printed characteristics (28) are left out, because the data actually used cannot be reconstructed.

Each two-class file, and each binarisation that a paper describes precisely enough to reproduce, defines one task: 113 tasks, used by 512 entries in 41 papers. One-vs-rest tasks that we built from multiclass files (171) are not tasks any paper evaluated and are reported apart. Every minority example is labelled safe, borderline, rare or outlier from the classes of its five nearest neighbours [NAPIERALA2016TYPES], with distances measured by HVDM [WILSON1997IMPROVED]. The noise index of a task is the share of minority examples that are borderline or outliers; we report it as a continuous value because we found no published threshold for it. Three flags use published thresholds: small disjuncts when at least 20% of minority examples are rare and outliers when more than 20% are outliers [NAPIERALA2016TYPES], and small sample when there are fewer than ten minority examples per feature [JAIN2000STATISTICAL]. The imbalance ratio $N_{\mathrm{maj}}/N_{\min}$, computed from the files, is grouped into the bands 1.5 to 3, 3 to 9 and above 9 [FERNANDEZ2008STUDY] and is not used as a difficulty flag.

Table 4 shows that the most common property of the binary tasks is a small sample: 79 of 113 have fewer than ten minority examples per feature. The flags combine in only four profiles: small sample alone (55 tasks), small sample with outliers (13), all three flags (11) and none (34); every task with small disjuncts also carries the other two flags. The noise index separates tasks that the flags do not (Figure 1a): among the 34 tasks without a flag it ranges from 0 to 0.84, with ten tasks at 0.5 or above (haberman 0.84, bupa 0.79, vehicle1 0.74, yeast1 0.72 among them) and eight below 0.1. Its median is 0.37, 0.40, 0.31 and 0.50 in the imbalance-ratio bands up to 1.5 (4 tasks), 1.5 to 3 (22), 3 to 9 (19) and above 9 (68), and its rank correlation with the imbalance ratio is 0.15.

The measurable part of the corpus is also low-dimensional (Figure 1b). The printed number of features has a median of 9 and half of the entries lie between 4 and 10 features. The 59 entries with more than 100 features come from 54 names in 18 papers, no name is used by more than two

Table 4: Difficulty Profile of the Verifiable Tasks. A task can carry several flags; entries count how often the corpus uses a task.

	Binary and binarised		One-vs-rest
	Tasks	Entries	Tasks
All	113	512	171
Small disjuncts (rare $\geq 20\%$)	11 (10%)	39 (8%)	7 (4%)
Outliers ($> 20\%$)	24 (21%)	120 (23%)	14 (8%)
Small sample ($N_{\min}/d < 10$)	79 (70%)	345 (67%)	70 (41%)
None of the three	34 (30%)	167 (33%)	97 (57%)
Imbalance ratio > 9	68	284	90
Noise index, median [P25, P75]	0.40 [0.25, 0.72]		0.16 [0.05, 0.45]

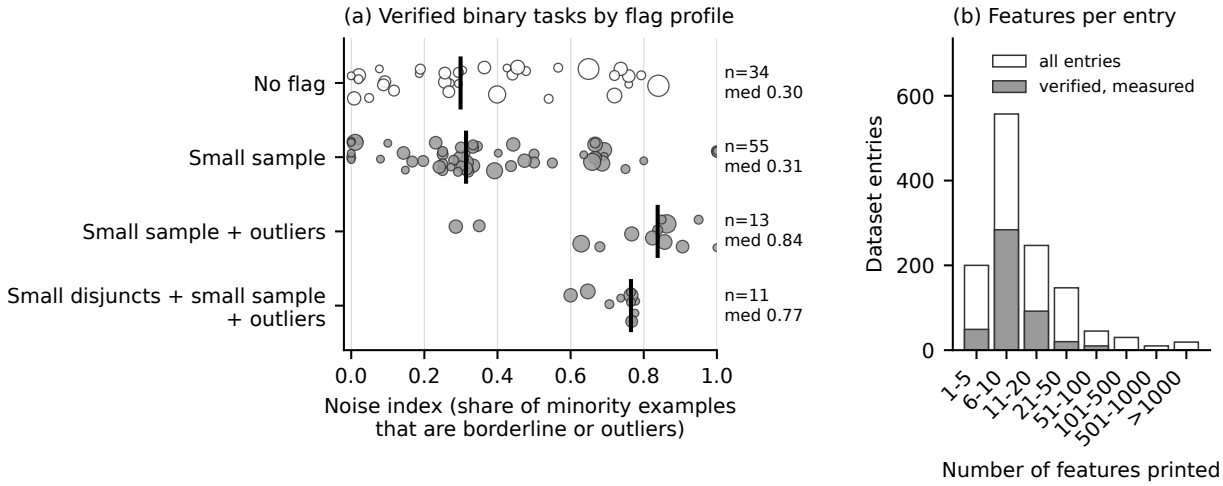


Figure 1: (a) Noise index of the 113 verifiable binary and binarised tasks, one row per flag profile; marker area grows with the number of entries that use the task, and the vertical bar marks the median. (b) Number of features printed for the 1,255 entries that report it, with the entries that could be verified and measured. The bins are for display only.

papers, and none of them could be verified; the verified tasks have at most 85 features (median 9).

The imbalance ratio, the one difficulty-related number most papers print, says little about where the minority examples lie: tasks with a low ratio can be as mixed as tasks with a high one, and a task without any flag is not necessarily easy. Conclusions drawn from the common benchmarks speak mainly for small, low-dimensional tables with few minority examples per feature.

Most papers mix these profiles. Of the 29 papers with at least three verifiable tasks, 26 include both flagged and flag-free tasks, two (SMOTE [CHAWLA2002SMOTE] and SMOTE-ENN) use only flag-free tasks and one (MLOS) only flagged ones. Section 1.6 relates the measured profiles to the reported wins.

A reference collection. For readers who need a small benchmark that covers the measured range, Table 5 selects 27 of the 113 tasks by a fixed rule with no threshold of our own. Each combination of flag profile and imbalance-ratio band that occurs (ten combinations) contributes the task that the corpus uses most, for comparability with earlier results, and the tasks with the

Table 5: A Reference Collection of Verifiable Two-Class Tasks. One row per selected task; Role: core = most used task in its cell, low/high = lowest/highest noise index in its cell. Profile: S small sample, O outliers, D small disjuncts; Noise: noise index; Entries: corpus entries that use the task.

Task	Role	Profile	IR band	N	$N_{\min}$	d	IR	Noise	Entries
australian	low	none	≤1.5	690	307	14	1.2	0.29	1
bupa	core+high	none	≤1.5	345	145	6	1.4	0.79	2
iris0	low	none	1.5–3	150	50	4	2.0	0.02	2
haberman	core+high	none	1.5–3	306	81	3	2.8	0.84	21
segment0	low	none	3–9	2,308	329	19	6.0	0.02	7
yeast3	core	none	3–9	1,484	163	8	8.1	0.40	13
adult	high	none	3–9	48,842	11,687	14	3.2	0.54	2
kr-vs-k-one_vs_fifteen	low	none	>9	2,244	78	6	27.8	0.00	1
shuttle-c0-vs-c4	core	none	>9	1,829	123	9	13.9	0.01	7
yeast-0-2-5-6_vs_3-7-8-9	high	none	>9	1,004	99	8	9.1	0.57	2
heart	core+low	S	≤1.5	270	120	13	1.2	0.33	4
sonar	high	S	≤1.5	208	97	60	1.1	0.40	1
wine [minority: 1]	low	S	1.5–3	178	59	13	2.0	0.00	1
glass0	core	S	1.5–3	214	70	9	2.1	0.44	7
glass1	high	S	1.5–3	214	76	9	1.8	0.47	7
ecoli2	low	S	3–9	336	52	7	5.5	0.23	6
ecoli3	core	S	3–9	336	35	7	8.6	0.69	10
hepatitis (PMLB, N=155)	high	S	3–9	155	32	19	3.8	0.75	2
dermatology-6	low	S	>9	358	20	34	16.9	0.00	2
yeast5	core	S	>9	1,484	44	8	32.7	0.66	13
cleveland-0_vs_4	high	S	>9	177	13	13	12.6	1.00	4
ecoli-0-1-3-7_vs_2-6	low	S+O	>9	281	7	7	39.1	0.29	7
yeast4	core	S+O	>9	1,484	51	8	28.1	0.86	15
winequality-red-3_vs_5	high	S+O	>9	691	10	11	68.1	1.00	1
yeast-1-4-5-8_vs_7	low	D+S+O	>9	693	30	8	22.1	0.60	7
glass2	core	D+S+O	>9	214	17	9	11.6	0.65	9
winequality-red-8_vs_6-7	high	D+S+O	>9	855	18	11	46.5	0.78	1

lowest and the highest noise index, for spread. The collection spans noise indices from 0 to 1, imbalance ratios from 1.1 to 68 and sizes from 150 to 48,842 instances, and 17 of its files match the official KEEL table. Its tasks are not independent of one another: it holds five yeast, three ecoli and three glass splits, which should be grouped by source when results are tested.

Limits of the measurement. The files come from mirrors of the repositories (the keel-ds and imbalanced-databases packages and PMLB); for the 72 KEEL files in the official KEEL table, instance and feature counts match in every case, and the remaining files have not yet been compared with the repositories. PMLB stores nominal attributes as integer codes, so they enter HVDM as numeric attributes. The labels are computed in the full feature space, not in any two-dimensional projection.

Evidence: difficulty/verifiability_entries.csv (643 verifiable entries, 61 papers, 131 files); difficulty/measure.py (HVDM, $k = 5$, Napierala types incl. the rule for rare examples) → difficulty_measures.csv; group_cited.py → difficulty_groups_cited.csv; profile_final.py → profile_final_summary.json (Table 4); fig_noise_summary.json (rank correlation); dimension_profile.json; keel_official_check.csv; per-paper profiles: paper_profile_mix.py → paper_profile_mix.json; reference collection: reference_collection.py → reference_collection.csv (rule and coverage in reference_collection_summary.json; independent check reference_collection_check.py); figure: difficulty/fig_manuscript.py; reported properties: out/characteristics_coverage.json

Table 6: Comparisons of the Proposed Method with Plain SMOTE by Metric (29 Papers). A row is one dataset and one metric in one paper; printed values are compared in the direction of the metric, equal printed values are ties, and n.c. counts values not printed as numbers. Scores are not comparable across papers.

Metric	Papers	Rows	Proposed wins	Ties	SMOTE wins	n.c.
AUC	19	403	309	30	64	0
G-mean	19	380	283	29	64	4
F-measure	18	287	217	16	50	4
Recall (sensitivity)	12	193	130	21	42	0
Precision	7	113	62	5	46	0
Accuracy	4	69	62	1	6	0
MCC	3	52	40	0	8	4
AUC-PR	1	7	3	0	0	4
Specificity	1	7	1	0	6	0
All metrics	29	1,511	1,107	102	286	16

(Analysis E).

1.6 Where Reported Wins over SMOTE Come From

At the paper level, 44 of the 73 papers use plain SMOTE as a baseline, 13 use SMOTE inside a multiclass or ensemble scheme, and 16 do not compare with SMOTE at all. Fewer papers give scores a reader can pair: 29 papers print per-dataset scores for the proposed method and plain SMOTE in the same table. For these papers we compared the printed values on every metric that the table reports for both methods, with the classifier and condition fixed as in Section 1.1; a win means the proposed method's printed value is better in the metric's direction, and equal printed values count as a tie. This gives 1,511 comparisons on 524 datasets (one to six metrics per paper, median three); 16 cannot be decided because a value is not printed as a number. Of the other 1,495 (520 datasets), the proposed method wins 1,107, ties 102 and loses 286, a win in 79.5% of the decided comparisons (95% CI 77.3 to 81.5). Twenty-three of the 29 papers report at least one comparison that SMOTE wins.

Table 6 shows that the margin is not the same on every metric. On AUC, G-mean, F-measure and recall the proposed method wins 76 to 83% of the decided comparisons, and on accuracy and MCC 91 and 83%; on precision the split is only 62 to 46, and on specificity SMOTE wins six of seven. Counting one result per dataset (the majority over its metrics) gives 409 wins, 35 ties and 76 losses on 520 datasets, and the earlier single-metric rule (G-mean, else AUC, else F-measure, else accuracy) gives 521 comparisons, in which the proposed method wins in 402 rows, ties in 39 and loses in 80.

We tested the hypothesis that motivated this analysis: that proposed methods win more often on data the authors modified or generated than on original data. The data do not support it (Table 7). Modified and synthetic data give a win in 75.7% of decided comparisons (371 of 490) against 79.3% on original data. Because the comparisons of one paper are not independent, we estimated the difference with the paper as the unit of clustering: a logistic model with paper-level clustering (generalised estimating equations) gives an odds ratio of 0.68 (95% CI 0.43 to 1.08, $p = 0.10$) for modified or synthetic against original data; a conditional logistic model that compares the two kinds of data only inside the 11 papers that use both and report both outcomes gives 0.61 (0.38 to 0.97, $p = 0.04$); and a mixed model with random effects for paper and for dataset gives

Table 7: Comparisons with Plain SMOTE by How the Dataset Was Obtained (All Metrics, 29 Papers). Win share is the share of wins among decided comparisons (ties left out), with a 95% Wilson interval. Sixteen comparisons not printed as numbers are left out.

How obtained	Datasets	Papers	Rows	Wins	Ties	Losses	Win share (%)
Original	144	25	425	329	10	86	79.3 (75.1–82.9)
Repository-defined	119	9	315	220	44	51	81.2 (76.1–85.4)
Modified	143	14	495	356	20	119	74.9 (70.9–78.6)
Synthetic	5	1	15	15	0	0	100 (79.6–100)
Cannot be told	105	13	234	184	24	26	87.6 (82.5–91.4)
Not matched to a dataset	4	3	11	3	4	4	42.9 (15.8–75.0)
All	520	29	1,495	1,107	102	286	79.5 (77.3–81.5)

0.75 (0.58 to 0.97). The direction is therefore the opposite of the hypothesis, but it is not stable: with one result per dataset the two clustered estimates are 0.64 ($p = 0.15$) and 0.46 ($p = 0.13$), and under the single-metric rule 0.80 ($p = 0.47$) and 0.61 ($p = 0.31$). Paper by paper, among the 13 papers with both kinds of data, the win share is higher on modified or synthetic data in 3, higher on standard data in 7 and equal in 3 (sign test $p = 0.34$). Repository-defined splits do not differ from original data (odds ratio 1.19, $p = 0.61$), and the comparisons whose data origin cannot be told show the highest raw win share apart from the single paper with synthetic data, a difference that disappears once papers are taken into account (odds ratio 0.99, $p = 0.98$).

The second hypothesis concerned papers that add a single standard dataset to data of another kind. Two papers fit this pattern: MWMOTE (one standard and 19 other datasets) and RUSBoost [seiffert2010rusboost] (one and 14). Both win on their standard dataset; on their modified datasets and those of unstated origin they win 52 of 65 decided comparisons (80.0%; two MWMOTE datasets that could not be matched to a record are left out), the same share as on such data in the 21 papers that have at least two standard datasets and also report other data (482 of 600, 80.3%). Five papers compare with SMOTE only on standard data and one (INOS) only on modified data. Two papers are too few for a test, and nothing in them suggests that the standard dataset serves a different purpose from the rest.

Measured difficulty says more about where the wins fall than the origin of the data. For the 72 verifiable two-class files (Section 1.5) that appear in these comparisons (20 papers, 374 comparisons), the chance that the proposed method beats SMOTE rises with the noise index: the odds ratio per 0.1 of noise index is 1.15 (95% CI 1.04 to 1.26, $p = 0.005$) with paper-level clustering and 1.19 (1.04 to 1.35, $p = 0.01$) within papers, and 1.16 (0.98 to 1.36, $p = 0.08$) with one result per file and paper. The imbalance ratio shows no such relation (odds ratio 0.92 per doubling, $p = 0.61$), nor do the small-sample, outlier and small-disjunct flags taken one at a time. This analysis is exploratory: it covers the files that could be verified, which are mostly small and low-dimensional.

Comparisons with SMOTE inside another scheme are kept apart and are not pooled with Tables 6 and 7. Twelve papers contribute 558 comparisons: Static-SMOTE (EMDO [yao2021emdo], OVO-SMOTE, SOUP) 79, with 69 wins, 3 ties and 7 losses; SMOTE after one-versus-one decomposition (AMDO [yang2018amdo], GDHS [yan2025gdhs], SHSampler [zheng2022shsampler]) 148, with 101, 6 and 41; SMOTE applied to all classes (MC-CCR [koziarski2020mcccr], MC-RBO [krawczyk2020mcrbo], NROMM) 239, with 231, 0 and 8; and SMOTE combined with another learner (DSRBF [fernandeznavarro2011dsrbf], EasyEnsemble, GAMO) 92, with 71, 2 and 19. Three papers are left out of every comparison set. KNNOR [islam2022knnor]

Table 8: Granularity of the Results Each Paper Reports (73 Papers)

Per-dataset table	47
Per-dataset, partial (some datasets, metrics or classifiers only)	8
Plots only	5
Averages over all datasets only	5
Ranks only	3
No comparison with other methods	5

prints only means over its 19 datasets (G-score .8363 for KNNOR and .8238 for SMOTE). MC-MBRC [MA2024MCMBRC] names SMOTE as a baseline, but the column headers of its per-dataset tables do not say which column is SMOTE. In RBBoost [DIEZPASTOR2015RANDOMBALANCE], the scores of its RB-Boost come from the Boosting-family table of its supplement (Supplementary Table 11) and those of its SMOTE ensemble (ESM) from the data-processing-family table (Supplementary Table 3); both report the same protocol (J48 with Laplace smoothing, ensembles of 100, stratified 5×2-fold cross-validation), but the paper compares methods only within each family and never compares RB-Boost with ESM directly. Its 86 datasets, 63 of them repository-defined, are kept in our data for audit.

What the reader should do. Read a reported win over SMOTE together with the metric: precision and specificity often favour SMOTE even when the other metrics do not. Expect wins over SMOTE to be more frequent on data whose minority examples are borderline or isolated, and less informative on clean data. Whether the data were modified does not, on this evidence, inflate the win rate; whether they are described well enough to be identified matters more for the reader (Sections 1.4 and 1.7).

Evidence: Rows: out/winloss_allmetric_rows.csv (2,497 dataset-by-metric rows from ai-draft/outcome/allmetric/*.json, 44 papers; the single-metric values re-copied independently and identical on 839 of 839 rows, out/winloss_allmetric_crosscheck.csv); build/winloss_allmetric.py → out/winloss_allmetric_summary.json (plain, non-plain groups, exclusions R4, R6, R8; paper classes aligned with build/reconcile_counts.py on 29 Oct 2026). Data types: build/allmetric_links.py → out/allmetric_links.csv (2,496 of 2,497 rows linked to a record). Tests: build/hypotheses.py → out/hypotheses.json, out/hypotheses_paper_split.csv (statsmodels 0.15: GEE exchangeable, ConditionalLogit, BinomialBayesMixedGLM variational fit). Single-metric set: out/verified_outcomes.csv, out/reconcile_counts.json (521 rows, rules R4, R6, R8). Difficulty: build/difficulty_outcomes.py → out/difficulty_outcomes.json. RBBoost: supplement Table 3 pp.3–5 and Table 11 pp.17–18; protocol p.101 and family design p.102. MC-MBRC: SMOTE named in Appendix A.1 and Table 8 only; Tables 9–12, p.10–11.

1.7 Can the Reported Results Be Checked?

Of the 57 papers that use SMOTE as a baseline (44 plain, 13 wrapped), 13 give a reader no per-dataset pair of scores in the paper itself: the results are plots (Borderline-SMOTE [HAN2005BORDERLINE], Safe-Level-SMOTE [BUNKHUMPORNPAT2009SAFELEVEL], Cluster-SMOTE [CIESLAK2006CLUSTER], k-means SMOTE [DOUZAS2018KMEANS], GLOS [HAN2023GLOS]), averages (G-SMOTE [DOUZAS2019GSMOTE], CISO [YADAV2025CISO], SplitBal [SUN2015SPLITBAL], QC-SMOTE [UPMAN2026QCSMOTE]), or ranks (MDO, RBO [KOZIARSKI2019RBO], MC-NCLWO [TAO2025MCNCLWO]), and the review [HASSANAT2022STOP] compares synthetic points rather than classifiers. A fourteenth paper, KNNOR, prints its pair of scores only as a mean over its datasets.

Even where per-dataset tables exist, they do not always cover the datasets a paper lists. In the 43 papers with per-dataset comparisons with SMOTE, 119 of the 952 listed dataset entries (12.5%,

95% CI 10.5 to 14.8) have no row in the table that compares the proposed method with SMOTE, and 12 papers account for all of them, most of all SMOTE-IPF [saez2015smoteipf] (32 of 41) and Loyalty-SMOTE (30 of 59). The gap is largest for synthetic sets (38 of 47) and for entries whose origin cannot be told (44 of 166), and smallest for modified and repository-defined data (5 of 180 and 7 of 189). Some of these datasets appear in other results tables of the same paper; what is missing for them is the pair of scores against SMOTE.

Baseline results are also reused across papers. QC-SMOTE copies its baseline numbers from the VS-SMOTE paper [qiu2025vssmote], and every SMOTE-all value in MC-CCR's Table 6 is identical, to two decimals and on all 20 datasets, to the SMOTE-all column of MC-RBO's Table VI, from the same group and with the same C5.0 setup. The same pattern appears across research groups: the Baseline, SSMOTE, GCS, ABNC, OSMOTE and AMDO columns of EMDO's Table 5 [yao2021emdo] repeat, value for value on all nine shared datasets, the MAUC results of AMDO's Table 6 [yang2018amdo]; EMDO states that it uses the same datasets as AMDO but not that the baseline results were copied. Reusing a baseline is efficient, but unless it is stated, a reader cannot tell whether the comparison was run under identical conditions. Several of these papers place per-dataset results on GitHub or in material outside the paper. Where only ranks are given, the paper's own table can still be informative: in RBO's Table 2 (CART, no added noise) SMOTE has the better mean G-mean rank, and CISO reports itself statistically equivalent to SMOTE.

For these papers the headline claim cannot be falsified from what is published. This does not mean the claims are wrong; it means a reader cannot tell whether an average improvement comes from consistent gains or from large gains on a few datasets.

Evidence: out/full_summary.json granularity and per-paper notes; MC-RBO Table VI p.10 vs MC-CCR Table 6 p.10 (ai-draft/full/MC-RBO.json notes). Listed but not compared: build/hypotheses.py → out/h3_entries.csv and out/hypotheses.json (H3b; dataset records linked to the fixed-metric and all-metric comparison rows). MWMOTE Table 5 audit: of 15 checkable rows, minority counts exact in 13, four rows with wrong N or majority (single paper, case only).

1.8 Recommendations for Reporting Data

1. Name the repository and version of every dataset, and whether the full set or a training file was used (Section 1.4).

2. State which class or classes form the minority and why, in the text and not only in a table (Section 1.3).

3. Declare every modification (binarisation, class merging, undersampling, noise, dropped attributes) in the data section, and print the size and class counts of the data actually used (Section 1.3).

4. Give the imbalance-ratio formula, not only its values (Section 1.4).

5. Report the number of source families alongside the number of datasets, and keep splits of one source together when testing for significance (Section 1.2).

6. Report per-dataset results for the proposed method and plain SMOTE on every metric used, with standard and modified datasets shown separately (Sections 1.6 and 1.7).

7. Separate tuning datasets from evaluation datasets, as CISO and LS [cieslak2008ls] do, and run the baselines rather than copying their numbers (Section 1.7).

8. Report at least one measure of where the minority examples lie (for example the shares of safe, borderline, rare and outlier examples) next to the imbalance ratio (Sections 1.5 and 1.6).